MISMERET

The Autonomous Operations Brain for Medicine Outside Hospital Walls

VentureHack 4 · i4Valley + GMC Playground · Shimi Network

Remote Patient Monitoring (7) · Prediction & Prevention (8) · Hospital Ops (5)

Runs today, in production: live demo, autonomous actions, tamper-proof ledger

The Problem

1. Who watches the screen at 3 AM? In the ICU: 942 alarms/day, one every 90 sec.
   85-99% of continuous-monitoring alarms have no clinical significance.
   At the patient's home there is no nurse at all - the problem is WORSE.

2. Coordination is still manual: phone, WhatsApp, Excel. US admin simplification
   potential: $270-320 BILLION per year (McKinsey).

3. Un-closed loops: a dying battery = 2 days of clinical blindness;
   a missed measurement = a monitoring disruption nobody saw.

The Solution: An Autonomous Operations Layer

1 **SENSE**
HL7 ADT · FHIR Subscriptions · provider webhooks. No vendor lock-in, no hardware.

2 **TRIAGE**
Suppress noise, rank severity, predict failure patterns (battery, adherence, equipment).

3 **ACT**
WhatsApp to patient, nurse visit scheduled, technician dispatched, systems updated - inside policy.

4 **VERIFY**
Every action checked against real outcome. Not 'sent' - 'done and verified'.

5 **LEDGER**
Tamper-proof audit log: billing base, learning base, regulatory asset.

Why This Is Real AI, Not a Chatbot

- Event-driven, not prompt-driven: no user asks - a stream of events decides, 24/7.
- Multi-agent hierarchy: sense, triage, act, verify are separate components with separate policies and accountabilities. Deterministic policy engine binds the probabilistic layer.
- Falsifiability-first: explicit failure thresholds defined before every action.
- Learning from outcomes: every action scored against the result it produced.
- Genuine engineering: inter-agent routing protocol, multi-tenant identity, fall-back chains, financial ledger with idempotency. Months of documented operational code in git.

Privacy & Regulation: Architectural, Not an Afterthought

Mishmeret never touches clinical content. The agent layer consumes only abstract operational events (battery state, measurement adherence, visit coordination).

- Data minimization: pseudonymized identifiers in the tamper-proof log.
- Patient contact only by consent, via verified business WhatsApp channels.
- In-country storage for Israeli institutions.
- Israeli Privacy Protection Law 5778-1998 + Ministry of Health guidelines; GDPR/HIPAA-ready architecture for abroad (BAA, encryption).

Clinical decisions always remain human. The system closes operational loops, not medical ones.

Unit Economics: Who Pays and How Much

Pilot program	200 home-hospitalized patients (assumption)
Coordinator hour	120 ILS/hour (sourced range 100-150)
Operational loops / patient / month	~6 (assumption)
Closed autonomously	55% assumption · pilot target 60-70%
Coordinator hours saved	~13 hours/week for the program
Billing model	Platform subscription + per verified closed action

The tamper-proof log makes billing fair and auditable: an autonomous, verified closed action = a real saved hour. Pay for verified outcomes, not promises.

Positioning: Complementary, Not Competing

RPM vendors = hardware + dashboard
Managed ops = humans in shifts
MISMERET = autonomous agent layer above any existing deployment, provider-agnostic, with a tamper-proof audit log.

It does not replace a single existing system - it operates the ones the customer already owns. That is also why integration complexity is LOW:
webhooks + existing messaging channels, no hardware, no medical-device pathway.

Scalability: One Brain, Many Domains

7 · Remote Monitoring	chronic care, home hospitalization
8 · Prediction & Prevention	sensor failure before it happens
5 · Hospital Operations	equipment & staff logistics
2/3/4 · Production · Supply · Energy	PdM $9-12B -> $42-69B · downtime >$1.4T/yr

Proven in our own production: the same architecture runs device telemetry,
GPU fleets and service lines today.

Market & The GMC Pilot Ask

RPM market: $26.6B (2024) -> $110.7B by 2033, CAGR 20% (Grand View)
US home healthcare: $61.5B (2026) -> $122.9B (2035)
Acute Care at Home: $6.1B -> $21.5B by 2034

Entry point: Israeli 'hospital-at-home' programs - GMC Playground: a live hospital with 650,000 patients in the Western Galilee. Pilot ask (90 days, measured from our own log):

60-70% loops closed with zero human touch

< 15 minutes from event to verified closure

20+ coordinator hours saved per week

Shimi Network · Live Today

A working AI-agent organization: multi-agent hierarchy, event-driven operations, inter-agent routing, tamper-proof financial ledger - running in production for months, documented and open to audit.

The submission itself - document, deck, demo - was built, critiqued, verified and upgraded BY this system. The network is the proof of the product.

We do not promise; we show the system running.

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